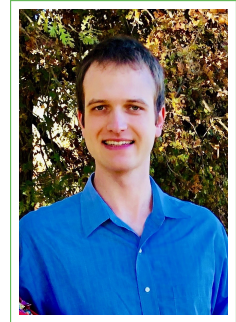


Luke Serafin

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Education

- 08/2019– **Ph.D. in Mathematics**, *Cornell University*, Ithaca, NY
Advanced to candidacy 09/2023. Note: Leave of absence Fall 2020–Spring 2022
- 08/2022–05/2024 **M.S. in Computer Science**, *Cornell University*, Ithaca, NY
- 08/2012–12/2015 **M.S. in Mathematical Sciences**, *Carnegie Mellon University*, Pittsburgh, PA
Note: Concurrent with bachelor's study
- 08/2011–05/2015 **B.S. in Mathematical Sciences with a double major in Logic & Computation**, *Carnegie Mellon University*, Pittsburgh, PA

Publications

- 07/2026 “Intergenerational Equity Without Nonprincipal Ultrafilters over $\mathbf{N}$.” Joint with Azul Fatalini. *In preparation*.
- 07/2026 “Equational Reasoning in ∞ -Categories.” Joint with Liron Cohen, Keri D’Angelo, and Dexter Kozen. *In preparation*.
- 04/2026 “Projective Chromatic Numbers.” Joint with Adrian Rettich. *Submitted*. arXiv:2604.21813
- 03/2026 “Funicular Preorders Can be Prelinearized Without Nonprincipal Ultrafilters Over $\mathbf{N}$.” Joint with Azul Fatalini. *Submitted*. arXiv:2602.13819
- 02/2026 “On Strongly Equitable Social Welfare Orders Without the Axiom of Choice.” *Submitted*. arXiv:2406.08684
- 02/2026 “Strong Morita Equivalence for Kleene Algebras and Related Structures.” Fahrenberg, Uli, Wesley Fussner, and Luigi Santocanale, eds. *Relational and Algebraic Methods in Computer Science: 22nd International Conference, RAMiCS 2026, Bedlewo, Poland, April 7–10, 2026, Proceedings*. Pp. 301–317. Springer Nature, 2026.
- 01/2026 “Ultrafilters over Successor Cardinals and the Tukey Order.” Joint with Tom Benhamou and Justin Moore. Accepted by *Fundamenta Mathematicae*. arXiv:2507.22307.
- 12/2017 “A Formally Verified Proof of the Central Limit Theorem.” Joint with Jeremy Avigad and Johannes Hölzl. *Journal of Automated Reasoning*, 59(4):389–423, 2017.
- 05/2016 “On Generalizations of Separating and Splitting Families.” Joint with Daniel Condon, Samuel Coskey, and Cody Stockdale. *The Electronic Journal of Combinatorics*, 23(3):3–36, 2016.

- 12/2015 “A Formally Verified Proof of the Central Limit Theorem.” Master’s thesis, Carnegie Mellon University, 2015. Advised by Professor Jeremy Avigad.
- 04/2015 “Mathematical Metaphysics.” Honour’s thesis, Carnegie Mellon University, 2015. Advised by Professor Clark Glymour.

Invited Presentations

- 05/15/2026 **“Prelinearization over the Solovay Model with Applications to Economics”**, *Set Theory in the United Kingdom, following participation in the workshop Research on Better Understanding of Solovay-Type Models*, Leeds, England
- 04/07/2026 **“Strong Morita Rigidity for Kleene Algebras”**, *Relational and Algebraic Methods in Computer Science*, Bedlewo, Poland
- 01/09/2026 **“Degrees of Impossibility in Social Choice”**, *Interdisciplinary seminar*, Arizona State University, Phoenix, AZ
Hosted by Sheila Miller-Edwards
- 11/21/2025 **“Morita Equivalence for Kleene Algebra”**, *Logic and Computation Seminar*, University of Pennsylvania, Philadelphia, PA
Hosted by Andre Scedrov
- 09/24/2025 **“Kleene Algebras and Morita Equivalence”**, *Logic and Applications (LAP)*, Dubrovnik, Croatia
- 09/17/2025 **“Tukey Types of Ultrafilters Over ω_1 ”**, *Logic Seminar*, Czech National Academy of Sciences, Prague, Czech Republic
Hosted by Christopher Lambie-Hanson
- 09/02/2025 **“Prelinearization, Social Welfare, and the Axiom of Choice”**, *Logic Seminar*, Carnegie Mellon University, Pittsburgh, PA
Hosted by Sumun Iyer
- 03/02/2025 **“Concerning Prelinearization of Analytic Preorders and Economics”**, *South Eastern Logic Symposium (SEALS)*, Gainesville, FL
- 02/13/2025 **“On Winning the Infinite Hat Game”**, *Mathematics Colloquium*, Miami University, Oxford, OH
Hosted by Paul Larson
- 02/10/2025 **“Choice and Equitable Social Welfare”**, *Logic Seminar*, Rutgers University, New Brunswick, NJ
Hosted by Filippo Calderoni
- 03/27/2024 **“Some Social Welfare Without Free Ultrafilters”**, *Logic Seminar*, University of Toronto, Toronto, ON
Hosted by Spencer Unger

Employment

- 08/2022– **Teaching Assistant, Research Assistant, and Grader**, Cornell University, Ithaca, NY
Semester-by-semester contracts with stipend to teach, carry out research, or provide grading support for courses including Theoretical Linear Algebra and Calculus (a year-long honours course), Differential Equations for Engineers, Calculus II, and Manifolds.

- 12/2020–02/2025 **Quality Assurance Lead Engineer**, *Sogeti USA*, Scottsdale, AZ
 Leading teams to design, implement, and maintain automated software testing for a large healthcare client.
 Note: Part-time beginning 08/2022 due to PhD.
- 05/2018–08/2019 **Software Engineer**, *Honeywell International*, Phoenix, AZ
 Designing, implementing, and maintaining safety-sensitive FAA-regulated software for the flight management systems of large airliners.
- 02/2016–12/2017 **Software Developer**, *Epic Systems Corporation*, Madison, WI
 Designing, implementing, and maintaining software to process and store healthcare provider records.
- 01/2012–12/2015 **Teaching Assistant, Research Assistant, Tutor, Grader, and Disability Resource Assistant**,
Carnegie Mellon University, Pittsburgh, PA
 Semester-by-semester undergraduate contracts to teach, carry out research, or provide grading or disability resource support for a course.
- 06/2014–08/2014 **Student Researcher**, *Boise State University*, Boise, ID
 Participant in 2014 Complexity Across Disciplines REU, with participation resulting in the 2016 publication of the paper “On Generalizations of Separating and Splitting Families” in the journal *Electronic Journal of Combinatorics* (23(3):3–36).

Honours

- Spring 2025 Phi Kappa Phi
 Fall 2014 Phi Beta Kappa

Technical Skills

- C programming language Academic and professional projects. Includes knowledge of C#, C++, and Java.
- SML Academic automated reasoning projects.
- Isabelle Formalization of Central Limit Theorem.
- L^AT_EX Preparation of academic papers.
- HTML, CSS, JavaScript, TypeScript Web development.
- Python Scripting for professional and academic projects.
- ADA Implementation of safety-sensitive software for airliners
- MUMPS Implementation and query interface for medical databases
- Ruby Some exposure as a scripting language for UI testing
- Lisp Some exposure as an implementation and interface language for automated theorem proving

Languages

- English (Native speaker)
- German (B2)
- French (B1)
- Italian (A2)
- Spanish (A2)
- Russian (A1)

- Hungarian (A1)
- Czech (A1)
- Polish (A1)